Entropy and space complexity of random approximate sets

Alexander Towell
atowell@siue.edu

2026

Abstract

We derive the joint entropy of the false positive and false negative error counts in the Bernoulli set model—a compositionally closed framework for random approximate sets with independent, element-wise errors. The joint entropy decomposes additively via element independence and admits a closed-form asymptotic expansion based on the binomial entropy approximation. We then establish the information-theoretic space lower bound for positive approximate sets over countably infinite universes, restating and proving the classical result of Carter, Floyd, Gill, Markowsky, and Wegman in the notation of the Bernoulli framework. The Bloom filter achieves $\ln 2 \approx 69\%$ of the optimal density. The general case—approximate sets with both false positives and false negatives ($0 < \tau < 1$)—remains an open problem.

Contents

1 Introduction **1**

2 Preliminaries **2**

3 Entropy of approximate sets **2**
3.1 Positives and negatives 3

4 Space complexity **3**

5 Conclusion **5**

1 Introduction

The Bernoulli set model [7] provides a compositional framework for random approximate sets—data structures whose membership queries may err with known false positive and false negative rates. A natural information-theoretic question is: how much uncertainty do these errors carry?

Motivation. The joint entropy of error counts quantifies the inherent unpredictability of a Bernoulli set realization. This entropy has direct implications for coding: any lossless representation of the error pattern requires at least this many bits on average. The space complexity question—how many bits are needed to implement an approximate set—connects the abstract Bernoulli model to the concrete cost of probabilistic data structures.

Contribution. We make two contributions. First, we derive the joint entropy of the false positive and false negative counts (theorem 3.1), showing that it decomposes additively by element independence and admits the closed-form asymptotic expansion

$$\log_2(2\pi e) + \frac{1}{2}\log_2\Big((u-m)\cdot m\cdot\varepsilon(1-\varepsilon)\cdot\omega(1-\omega)\Big) + \mathcal{O}\left(\frac{u}{m(u-m)}\right). \tag{1.1}$$

We extend this to the case where the number of positives is itself uncertain (section 3.1). Second, we restate and prove the classical space lower bound of Carter et al. [2] for positive approximate sets (proposition 4.1): any implementation over a countably infinite universe requires at least $-\log_2\varepsilon$ bits per element. We compare this to the Bloom filter's $-(1/\ln 2)\log_2\varepsilon$ bits per element and note that the general case $0 < \tau < 1$ remains open (section 4).

Related work. Shannon [4] introduced the entropy of binary channels; Cover and Thomas [3] provide the binomial entropy expansion we use. Carter et al. [2] established the space lower bound. Bloom [1] introduced the eponymous filter. The Bernoulli set model and its compositional algebra are developed in a companion paper [7].

Organization. Section 2 restates the Bernoulli axioms. Section 3 derives the joint entropy. Section 4 establishes the space complexity lower bound. Section 5 concludes with open problems.

2 Preliminaries

We restate the elements of the Bernoulli set model [7] used in this paper.

Axiom 1 (Element-wise independence)**.** *For all distinct $x, y \in U$, the error events at x and y are mutually independent:*

$$P(\mathbf{1}_{\tilde{A}}(x) \neq \mathbf{1}_A(x) \mid \mathbf{1}_{\tilde{A}}(y) \neq \mathbf{1}_A(y)) = P(\mathbf{1}_{\tilde{A}}(x) \neq \mathbf{1}_A(x)). \tag{2.1}$$

More generally, the family $\{\mathbf{1}_{\tilde{A}}(x) : x \in U\}$ is mutually independent given the objective set A and the error rates. Furthermore, within each partition block B_i, the error indicators $\{\mathbf{1}_{\tilde{A}}(x) : x \in B_i\}$ are identically distributed.

Axiom 2 (Conditional independence of block error rates)**.** *Given the objective set R, the random error rates $\alpha_1, \ldots, \alpha_n$ for the respective partition blocks $B_1, \ldots, B_n$ are mutually independent.*

From these axioms, the false positive count $\mathrm{FP}_m \sim \mathrm{Bin}(u-m, \varepsilon)$ and the false negative count $\mathrm{FN}_m \sim \mathrm{Bin}(m, \omega)$ are conditionally independent given m positives in a universe of u elements [7].

3 Entropy of approximate sets

The Bernoulli set model assigns binomial distributions to the number of false positives and false negatives. A natural question is how much *uncertainty* these error counts carry, as measured by Shannon entropy [4]. In this section, we derive the joint entropy of the false positive and false negative counts and extend the analysis to the case in which the number of positives is itself uncertain.

Theorem 3.1 (Joint entropy of false positives and false negatives)**.** *Given m positives in a universe of u elements, the joint entropy of the uncertain number of false positives and false negatives in a Bernoulli set with expected false positive rate ε and expected false negative rate ω is*

$$\mathcal{H}(\mathrm{FP}_m, \mathrm{FN}_m) = \log_2(2\pi e) + \frac{1}{2}\log_2\Big((u-m)\cdot m\cdot\varepsilon(1-\varepsilon)\cdot\omega(1-\omega)\Big) + \mathcal{O}\left(\frac{u}{m(u-m)}\right), \quad (3.1)$$

where the asymptotic term holds as $u \to \infty$ with m/u bounded away from 0 and 1 (so that both m and $u-m$ grow with u).

Proof. By axiom 1, the random error counts FP_m and FN_m are conditionally independent given m positives (they are functions of indicators on disjoint element sets: negatives and positives respectively). Therefore the joint entropy decomposes additively:

$$\mathcal{H}(\mathrm{FP}_m, \mathrm{FN}_m) = \mathcal{H}(\mathrm{FP}_m) + \mathcal{H}(\mathrm{FN}_m). \quad (a)$$

The random variable FP_m is binomially distributed, $\mathrm{FP}_m \sim \mathrm{Bin}(u-m, \varepsilon)$. The entropy of a binomial distribution $\mathrm{Bin}(n,p)$ admits the well-known asymptotic expansion [3]

$$H = \frac{1}{2}\log_2\Big(2\pi e\cdot n\cdot p(1-p)\Big) + \mathcal{O}(1/n). \quad (b)$$

Applying eq. (b) to FP_m and FN_m respectively yields

$$\mathcal{H}(\mathrm{FP}_m) = \frac{1}{2}\log_2(2\pi e) + \frac{1}{2}\log_2\Big((u-m)\,\varepsilon(1-\varepsilon)\Big) + \mathcal{O}\left(\frac{1}{u-m}\right), \quad (c)$$

$$\mathcal{H}(\mathrm{FN}_m) = \frac{1}{2}\log_2(2\pi e) + \frac{1}{2}\log_2\Big(m\,\omega(1-\omega)\Big) + \mathcal{O}\left(\frac{1}{m}\right). \quad (d)$$

Summing these expressions and combining the logarithms gives

$$\mathcal{H}(\mathrm{FP}_m, \mathrm{FN}_m) = \log_2(2\pi e) + \frac{1}{2}\log_2\Big((u-m)\,m\,\varepsilon(1-\varepsilon)\,\omega(1-\omega)\Big) + \mathcal{O}\left(\frac{u}{m(u-m)}\right), \quad (e)$$

where the error term follows from $\mathcal{O}(1/(u-m)) + \mathcal{O}(1/m) = \mathcal{O}\Big(u/(m(u-m))\Big)$. □

3.1 Positives and negatives

When the number of positives is itself uncertain, modeled by the random variable P with probability mass function $\mathrm{p_P}(p|u)$, the joint distribution of positives, false positives, and false negatives factorizes as

$$\mathrm{p}(p, f_p, f_n | u, \varepsilon, \omega) = \mathrm{p_P}(p|u)\ \mathrm{p_{FP}}(f_p|p,u,\varepsilon)\ \mathrm{p_{FN}}(f_n|p,u,\omega). \quad (3.2)$$

By axiom 1, FP and FN are conditionally independent given P, so the joint entropy decomposes as

$$\mathcal{H}(\mathrm{P}, \mathrm{FP}, \mathrm{FN}) = \mathcal{H}(\mathrm{P}) + \sum_{p=0}^{u} \mathrm{p_P}(p|u)\ \mathcal{H}(\mathrm{FP}\mid p) + \sum_{p=0}^{u} \mathrm{p_P}(p|u)\ \mathcal{H}(\mathrm{FN}\mid p). \quad (3.3)$$

The last two sums are the conditional entropies $\mathcal{H}(\mathrm{FP}\mid\mathrm{P})$ and $\mathcal{H}(\mathrm{FN}\mid\mathrm{P})$, each weighted by the distribution of positives.

Example 1 *The* expected *number of false positives, marginalizing over* P*, is*

$$\begin{aligned}\mathrm{E}\left[\mathrm{FP}\right] &= \sum_{p=0}^{u} \mathrm{p_P}(p|u)\,\mathrm{E}\left[\mathrm{FP}_p\right] = \sum_{p=0}^{u} \mathrm{p_P}(p|u)\,(u-p)\,\varepsilon \\ &= \varepsilon\Big(u - \mathrm{E}\left[\mathrm{P}\right]\Big) = \varepsilon \cdot \mathrm{E}\left[\mathrm{N}\right], \end{aligned} \tag{a}$$

confirming that the marginal expected false positive count depends only on the expected number of negatives and the false positive rate ε.

4 Space complexity

If the finite cardinality of a universe is u and the set is *dense* (and the approximation is also dense, i.e., the false negative rate is relatively small), then

$$\mathcal{O}(u) \text{ bits} \tag{4.1}$$

are needed to code the set, which is independent of p, the false positive rate, and the false negative rate.

The lower-bound on the *expected* space complexity of a data structure that models the *positive random approximate set* where the elements are over a *countably infinite* universe is a classical result due to Carter, Floyd, Gill, Markowsky, and Wegman [2]. We restate it here in the notation of the present paper and include a proof for completeness.

Proposition 4.1 ([2])**.** *The* information-theoretic lower-bound *of a data structure that implements the countably infinite* positive random approximate set *abstract data type has an* expected *bit length given by*

$$-\log_2 \varepsilon \text{ bits/element}, \tag{4.2}$$

where $\varepsilon > 0$ is the false positive rate.

Proof. It suffices to prove the bound for deterministic encodings; the extension to randomized data structures follows from Yao's minimax principle.

Consider a finite subset of the universe with n elements, from which an objective set S of size p is drawn. A deterministic encoding using B bits maps each input set S to a state $r = f(S)$ from at most 2^B distinct states. Each state r determines a *positive response set* $Q(r) = \{x : \mathrm{query}(x, r) = 1\}$. Since the data structure has no false negatives ($\tau = 1$), we have $S \subseteq Q(r)$ for the state $r = f(S)$ encoding S. The false positive rate constraint requires that at most an ε-fraction of the $n - p$ negatives belong to $Q(f(S))$ for every input set S, so $|Q(f(S))| \leq p + \varepsilon(n-p)$ for all S. In particular, $|Q(r)| \leq p + \varepsilon(n-p)$ for every state r in the image of f.

By the pigeonhole principle, some state r^* must encode at least $\binom{n}{p}/2^B$ distinct input sets. Every such set is a subset of $Q(r^*)$ of size p, so

$$\binom{n}{p}/2^B \leq \binom{|Q(r^*)|}{p} \leq \binom{p + \varepsilon(n-p)}{p}. \tag{a}$$

Taking logarithms and rearranging:

$$B \geq \log_2 \binom{n}{p} - \log_2 \binom{p + \varepsilon(n-p)}{p}. \tag{b}$$

For fixed p as $n \to \infty$, $\binom{n}{p} \sim n^p/p!$ and $\binom{p+\varepsilon n}{p} \sim (\varepsilon n)^p/p!$, so

$$B \geq \log_2 \frac{n^p}{(\varepsilon n)^p} = p \log_2 \frac{1}{\varepsilon} = -p \log_2 \varepsilon, \tag{c}$$

giving $-\log_2 \varepsilon$ bits per element. □

Remark. *The general case $0 < \tau < 1$ remains open. When false negatives occur, $S \not\subseteq Q(r)$ and the pigeonhole step above fails: valid input sets are no longer constrained to be subsets of the positive response set. The heuristic scaling $-\tau \log_2 \varepsilon$ (retaining $p\tau$ elements on average, each requiring $-\log_2 \varepsilon$ bits) is plausible but is not a valid information-theoretic lower bound, since the encoding problem changes qualitatively when elements are randomly dropped.* △

A well-known implementation of countably infinite *positive approximate set* is the Bloom filter [1] which has an expected space complexity given by

$$-\frac{1}{\ln 2} \log_2 \varepsilon \text{ bits/element}, \tag{4.3}$$

thus the absolute efficiency of the Bloom filter is $\ln 2 \approx 0.69$. A practical implementation of the *positive random approximate set* is given by the *Perfect Hash Filter* [5], which compares favorably to the Bloom filter in many circumstances.[1]

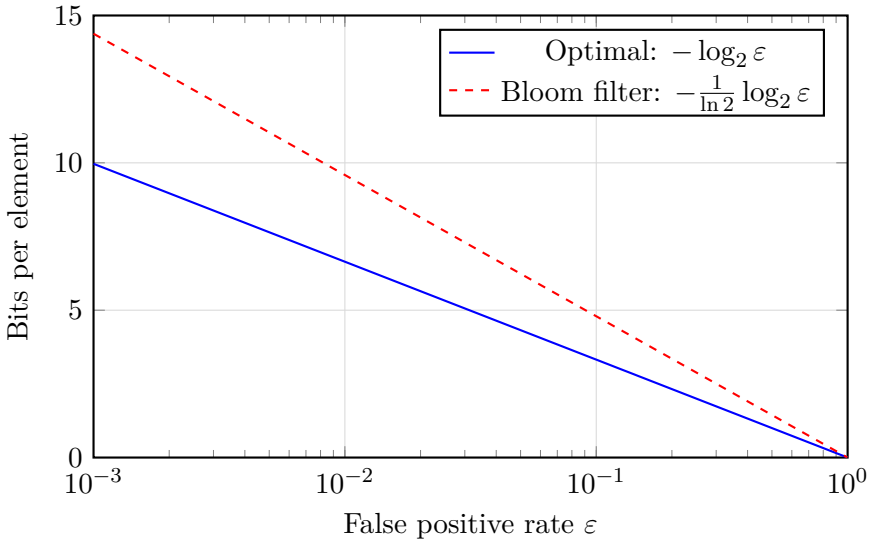


Figure 1: Space complexity of positive approximate sets as a function of the false positive rate ε. The solid curve is the information-theoretic lower bound $-\log_2 \varepsilon$ bits per element (proposition 4.1). The dashed curve is the Bloom filter cost $-(1/\ln 2) \log_2 \varepsilon$; the gap reflects the Bloom filter's $\ln 2 \approx 69\%$ efficiency.

5 Conclusion

We derived two information-theoretic results for the Bernoulli set model. First, the joint entropy of the false positive and false negative counts decomposes additively by element independence

[1] The *Singular Hash Set* [6] is an example of a data structure that obtains optimality using *brute-force* search, so it is not practical for even relatively small objective sets. However, its primary purpose is analytic tractability.

(theorem 3.1), yielding the closed-form asymptotic expansion

$$\mathcal{H}(\mathrm{FP}_m, \mathrm{FN}_m) = \log_2(2\pi e) + \frac{1}{2}\log_2\Big((u-m)\cdot m\cdot\varepsilon(1-\varepsilon)\cdot\omega(1-\omega)\Big) + \mathcal{O}\left(\frac{u}{m(u-m)}\right). \quad (5.1)$$

This result extends naturally to the case in which the number of positives is itself uncertain (section 3.1). Second, we restated and proved the classical information-theoretic space lower bound of Carter et al. [2] for positive approximate sets over countably infinite universes (proposition 4.1): at least $-\log_2\varepsilon$ bits per element are required. The Bloom filter [1] achieves $\ln 2 \approx 69\%$ of this optimal density.

Open problems. The information-theoretic space bound of proposition 4.1 assumes a positive approximate set ($\tau = 1$); the general case $0 < \tau < 1$ remains open, as noted in section 4. When false negatives occur, the pigeonhole argument fails because valid input sets are no longer constrained to be subsets of the positive response set. Establishing a tight lower bound for the general case is the most pressing open question.

The mutual information $I(\tilde{A}; A)$ between a Bernoulli set and its objective set, and the channel capacity of the element-wise binary channel viewed at the set level, remain unexplored.

Extending the entropy analysis to compositions of Bernoulli sets—where error rates are themselves functions of operand rates—would connect the present results to the compositional algebra of [8].

References

[1] Burton H. Bloom. Space/time trade-offs in hash coding with allowable errors. *Communications of the ACM*, 13(7):422–426, 1970.

[2] J. Lawrence Carter, Robert W. Floyd, John Gill, George Markowsky, and Mark N. Wegman. Exact and approximate membership testers. In *Proceedings of the 10th Annual ACM Symposium on Theory of Computing (STOC)*, pages 59–65, 1978.

[3] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2 edition, 2006.

[4] Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423, 1948.

[5] Alexander Towell. The perfect hash filter: an efficient, immutable implementation of the positive random approximate set abstract data type. Working paper. Available from the author upon request.

[6] Alexander Towell. The singular hash set. Working paper. Available from the author upon request.

[7] Alexander Towell. Bernoulli sets: a model for random approximate sets. Companion paper, 2026.

[8] Alexander Towell. Composition of Bernoulli sets. Companion paper, 2026.